

Dynamic Routing

Definition

- Dynamic routing is a technique in which **routers automatically learn, share, and update routes** to remote networks using **routing protocols**.
- Unlike **static routing**, routes are **not manually configured** — routers communicate with each other to build and maintain the routing table.
- It enables **automatic path selection** and **network adaptability** when topology changes occur.

Need for Dynamic Routing

- In large or complex networks, manually adding static routes is **time-consuming and error-prone**.
- Dynamic routing provides **automatic route discovery**, **link-failure recovery**, and **efficient load balancing**.

Characteristics

1. Routes are **learned and updated automatically**.
2. Uses **routing algorithms and protocols** to exchange information.
3. Periodic or triggered updates keep all routers synchronized.
4. Each router calculates the **best path** based on metrics (hop count, bandwidth, cost, delay, etc.).
5. The routing table is updated dynamically without manual intervention.

Types of Dynamic Routing Protocols

Type	Protocols	Description
Distance Vector	RIP, IGRP	Routers share routing tables with neighbors periodically. Uses hop count as metric.
Link State	OSPF, IS-IS	Routers exchange full topology information and compute shortest paths using algorithms like Dijkstra.
Hybrid	EIGRP	Combines distance-vector and link-state features; faster and more efficient.
Exterior Gateway	BGP	Used for routing between different autonomous systems (Internet level).

Advantages

- **Automatic updates** — no manual configuration needed.

- **Scalable** for large networks.
- **Adaptive** to topology or link failures.
- **Efficient path selection** based on metrics.
- **Load balancing** possible across multiple paths.

Disadvantages

- Consumes **more CPU, memory, and bandwidth** due to updates.
- Slightly **slower convergence** compared to static routing.
- **Complex configuration** for large protocols.
- Possibility of **routing loops** if timers and updates are mismanaged.

Examples of Dynamic Routing Protocols

Protocol	Type	Metric	Administrative Distance	Use Case
RIP (Routing Information Protocol)	Distance Vector	Hop Count	120	Small networks
OSPF (Open Shortest Path First)	Link State	Cost	110	Medium/Large enterprise
EIGRP (Enhanced Interior Gateway Routing Protocol)	Hybrid	Bandwidth & Delay	90	Cisco networks
BGP (Border Gateway Protocol)	Exterior Gateway	Path Attributes	20 (external)	ISP-to-ISP routing

Working Principle

1. Routers exchange **Hello** or **Update** messages to discover neighbors.
2. Each router builds a **topology database** of reachable networks.
3. A **routing algorithm** computes the best path (lowest metric).
4. The **routing table** is updated automatically.
5. If a route fails, routers recompute and propagate updates to others.

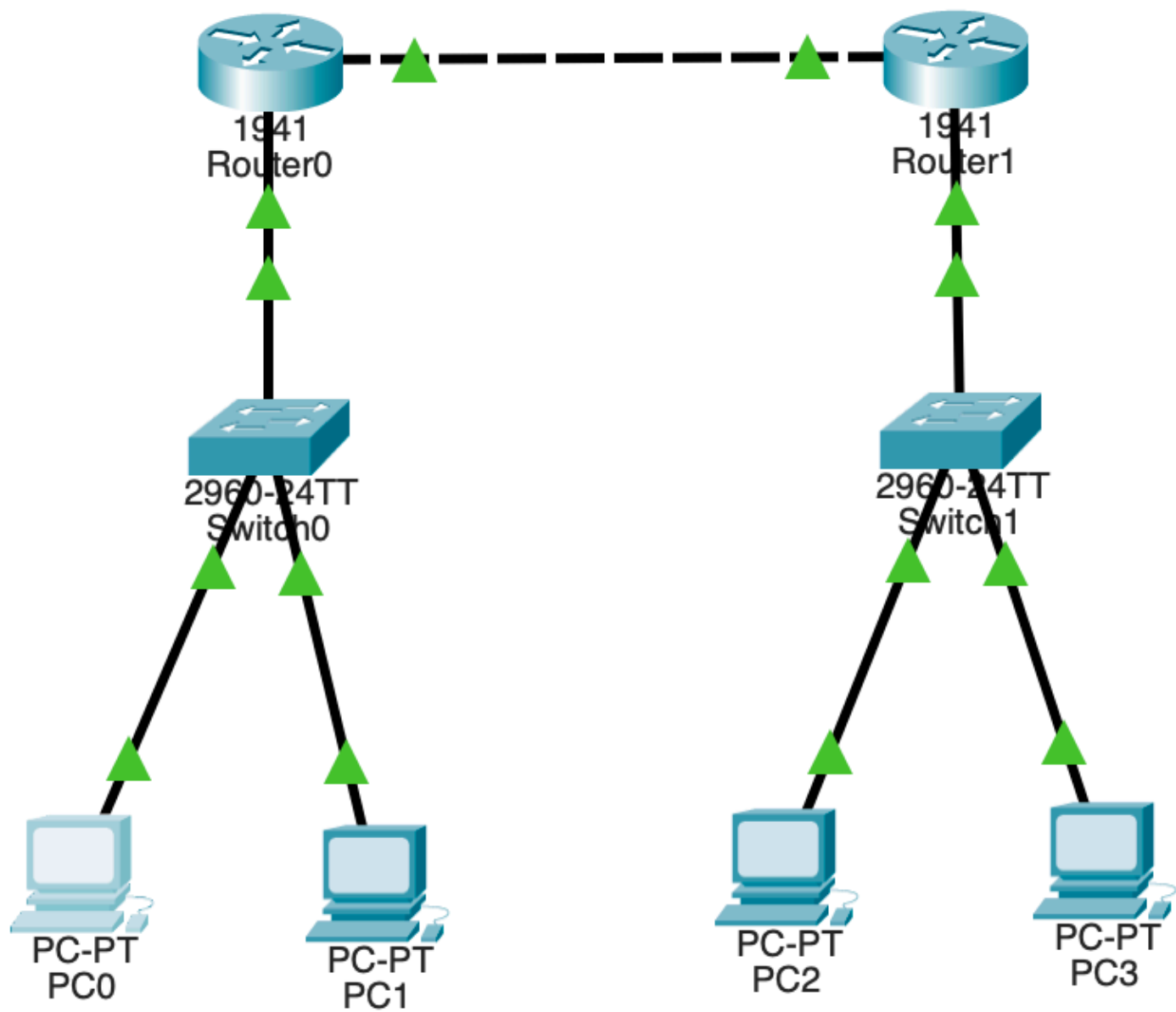
Dynamic Routing Configuration (RIP) – Cisco Packet Tracer

Apparatus

- 2 Routers (Cisco 1941)
- 2 Switches (Cisco 2960)

- 4 PCs
- Straight-through and crossover cables
- Cisco Packet Tracer software

Network Topology



IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	G0/0	192.168.1.1	255.255.255.0	-
Router0	G0/1	10.0.0.1	255.255.255.252	-
Router1	G0/0	192.168.2.1	255.255.255.0	-
Router1	G0/1	10.0.0.2	255.255.255.252	-
PC0	Fa0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	Fa0	192.168.1.3	255.255.255.0	192.168.1.1

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC2	Fa0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	Fa0	192.168.2.3	255.255.255.0	192.168.2.1

Procedure

Step 1: Device Connections

- Connect PCs to switches using straight-through cables.
- Connect each switch to its router using straight-through cables.
- Connect Router0 and Router1 using a crossover cable (G0/1 ↔ G0/1).

Step 2: Configure Interfaces

Router0

Interface G0/0 → IP 192.168.1.1 255.255.255.0
Interface G0/1 → IP 10.0.0.1 255.255.255.252
Turn both interfaces ON

Router1

Interface G0/0 → IP 192.168.2.1 255.255.255.0
Interface G0/1 → IP 10.0.0.2 255.255.255.252
Turn both interfaces ON

Step 3: Configure RIP (Dynamic Routing)

Router0 → Config → Routing → RIP

Add:
192.168.1.0
10.0.0.0

Router1 → Config → Routing → RIP

Add:
192.168.2.0
10.0.0.0

This enables both routers to exchange route information automatically.

Step 4: Configure PCs

Assign IPs and gateways as per the IP table above using Desktop → IP Configuration.

Step 5: Verification

1. On Routers (CLI):

```
show ip route
```

Expected Output:

- Router0: R 192.168.2.0 [120/1] via 10.0.0.2
- Router1: R 192.168.1.0 [120/1] via 10.0.0.1

2. On PC:

```
ping 192.168.2.2
```

Ping reply indicates successful routing.
